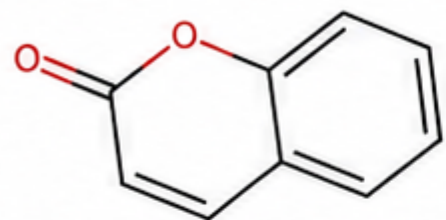
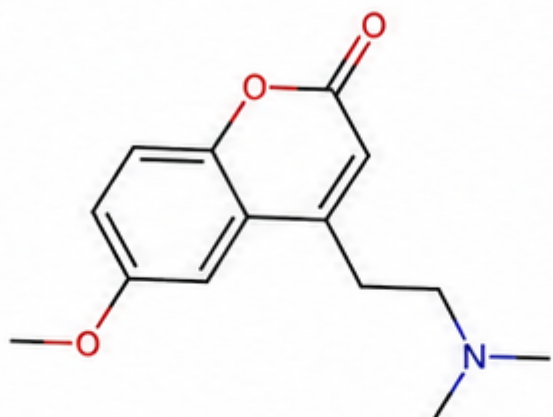


What persistent homology sees

Shared scaffold core
(real scaffold from top-20)



Side-chain-expanded analogue
(illustrative)



H1 (rings)
persists



H0 (connectivity)
diverges

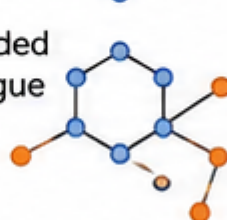


H1 = rings, conserved
H0 = connectivity, divergent

core

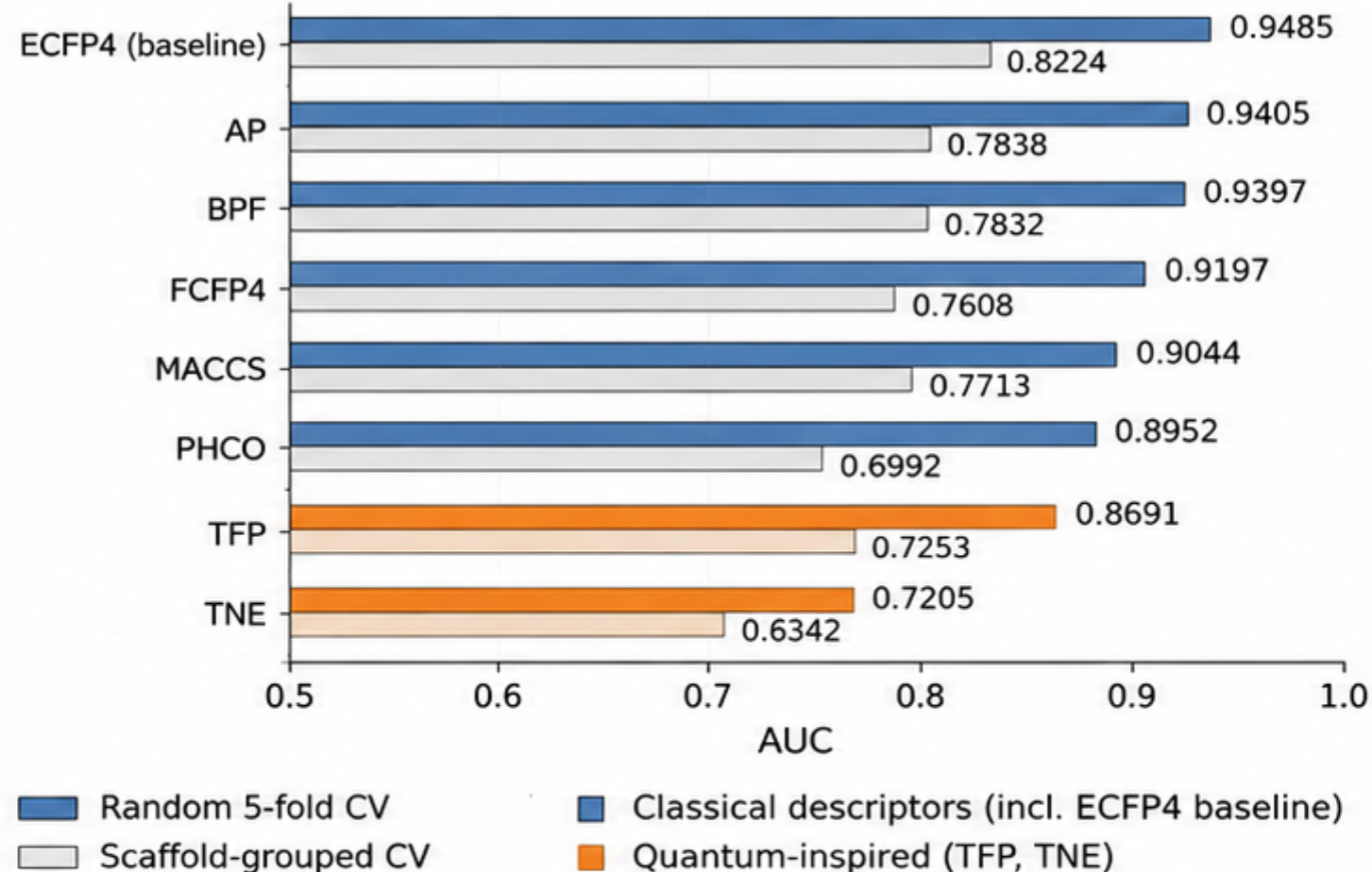


expanded
analogue



92.6% whole-molecule novelty, 69.3% scaffold retention
H1 conserved, H0 divergent

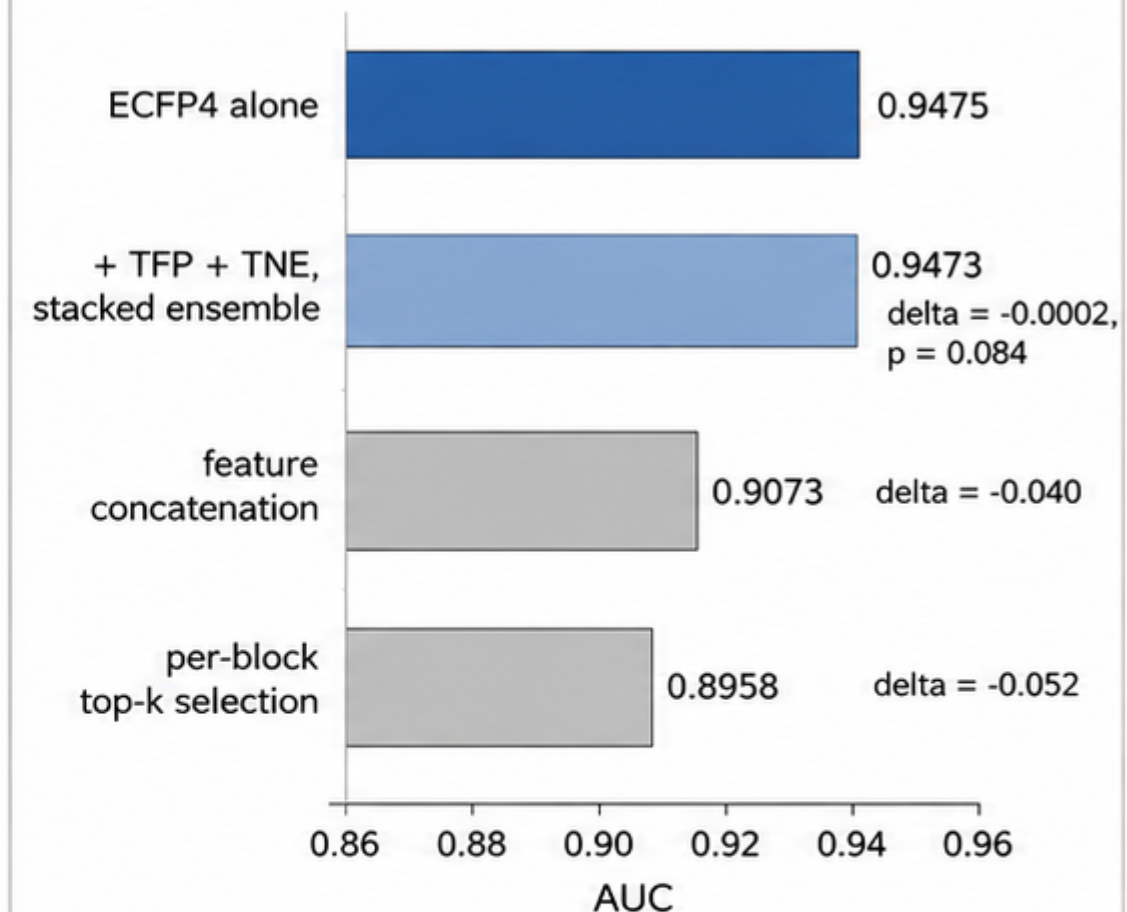
What random splits hide



**Every descriptor loses 0.09-0.20 AUC when
scaffolds cannot leak across folds**

632 Bemis-Murcko scaffolds across 19,836 molecules;
benzene alone is the core of 1,661 (8.4%)

What fusion adds



**Adding TFP+TNE to ECFP4:
no detectable gain (delta AUC = -0.0002, p = 0.084)**



Activity labels are computational predictions (Ersilia eos80ch model),
not experimental measurements.



Quantum kernel simulated on a classical statevector simulator -
no quantum hardware, no quantum advantage claimed.